

Initial-algebra Semantics

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Introduction

My object of study:

PROGRAMMING LANGUAGES

↳ build mathematical models

Goal  Generic method

Description of a language

$\mapsto$

Its model

Introduction

My object of study:

PROGRAMMING LANGUAGES

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Goal 🎯 Generic method

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⟶

Its model

↳ Skel

Design a notion of model

- suitable for Skel
- good mathematical properties

🤔 my last year seminar

🤔 this year's seminar?

Initial-algebra semantics

Goal  Get a **mathematical** representation/definition of a language or system
↳ e.g. prove and apply abstract theorems, study properties, etc.

Specify programming languages **implicitly**:

- first define a whole **category of models**,
- desired language implicitly defined as **initial object** therein

(Category theory – 1/4 Categories)

Definition (Category C)

- a collection $\text{ob } C$ of **objects**
- a collection $\text{mor } C$ of **morphisms** such that
 - each morphism $f \in \text{mor } C$ has a **source** A and a **target** B ($A, B \in \text{ob } C$)

$$f : A \rightarrow B$$

- morphisms **compose**: if $f : A \rightarrow B$ and $g : B \rightarrow C$,

$$g \circ f : A \rightarrow C$$

- there are **identities**: for any $A \in \text{ob } C$,

$$\text{id}_A : A \rightarrow A \quad \text{id}_A \circ f = f \quad f \circ \text{id}_A = f$$

Example

Sets and functions $\rightsquigarrow$ a category **Set**, groups and group homomorphisms $\rightsquigarrow$ **Grp**, any poset, ...

(Category theory – 2/4 Functors)

Definition (Functor)

C and D are categories, a **functor** $F : C \rightarrow D$ is

- a function on **objects** $F : \text{ob } C \rightarrow \text{ob } D$
- that extends to **morphisms**

$$F(f : A \rightarrow B) = F(f) : FA \rightarrow FB$$

- that **preserves identities and composition**

$$F(\text{id}_A) = \text{id}_{FA} \quad F(g \circ f) = Fg \circ Ff$$

Example

“Forgetful” functor $U : \mathbf{Grp} \rightarrow \mathbf{Set}, \dots$

Proposition

Categories and functors $\rightsquigarrow$ a category **Cat**

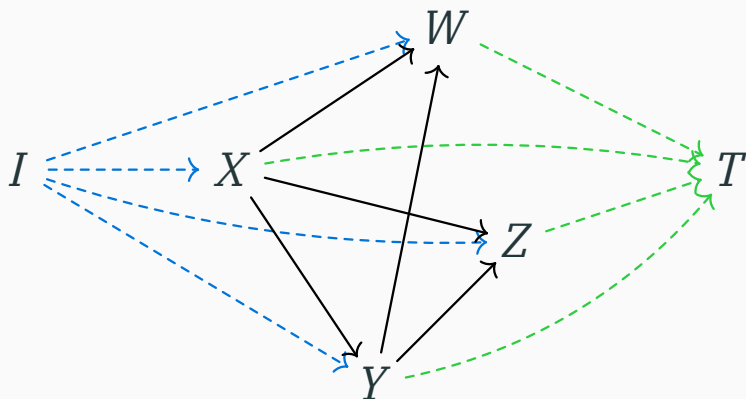
(Category theory – 3/4 Some limits and colimits)

Definition (Initial object)

Object I is **initial** in C iff $\forall X \in \text{ob } C, \exists ! f : I \rightarrow X \in \text{mor } C \quad \rightsquigarrow$ “smallest” object

Definition (Terminal object)

Object T is **terminal** in C iff $\forall x \in \text{ob } C, \exists ! f : X \rightarrow T \in \text{mor } C \quad \rightsquigarrow$ “biggest” object



Example

In a poset,

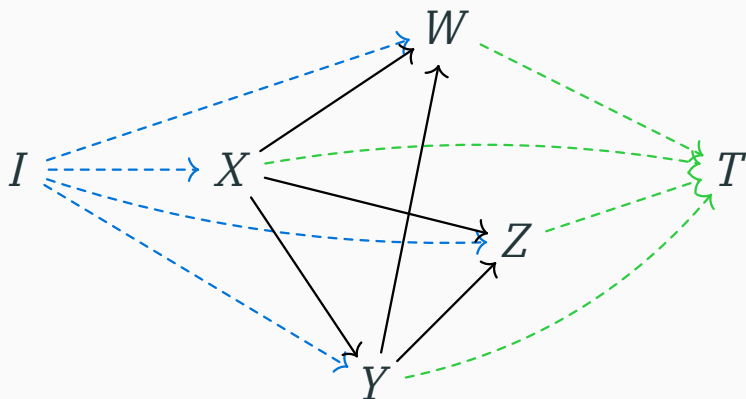
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Example

In a poset, $\perp$ is initial and $\top$ is terminal.

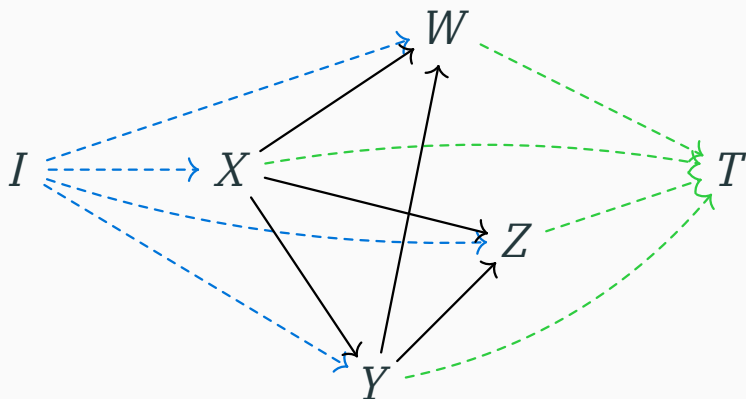
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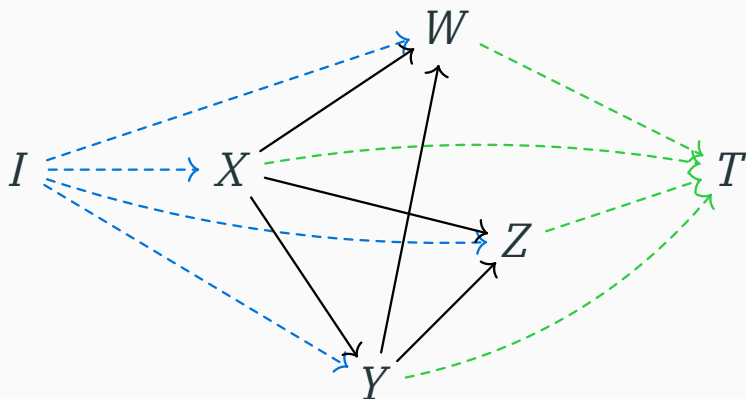
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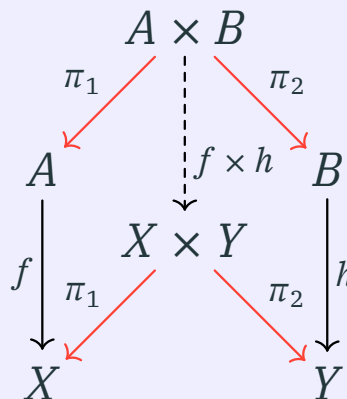
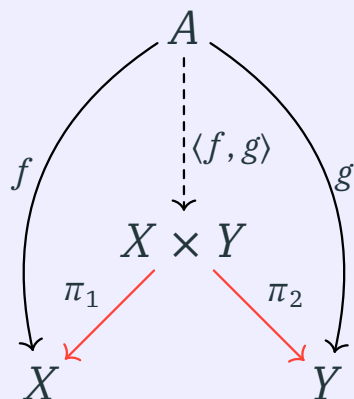
In a poset, $\perp$ is initial and $\top$ is terminal.

In **Set**, $\emptyset$ is initial and $\{*\}$ is terminal.

(Category theory – 4/4 Some limits and colimits)

Definition (Product)

A **product** of $X, Y \in \text{ob } C$ is an object $X \times Y \in \text{ob } C$ such that



projection morphisms

- $\pi_1 : X \times Y \rightarrow X$
- $\pi_2 : X \times Y \rightarrow Y$

$$\hookrightarrow \Delta_X = \langle \text{id}_X, \text{id}_X \rangle : X \rightarrow X \times X$$

Example

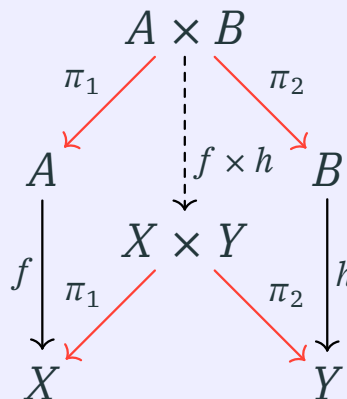
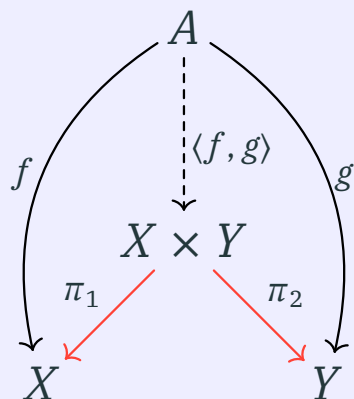
In **Set**, cartesian product.

$$\langle f, g \rangle(a) = (f(a), g(a)) \quad \Delta_X(x) =$$

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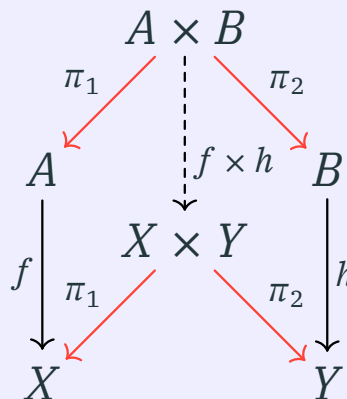
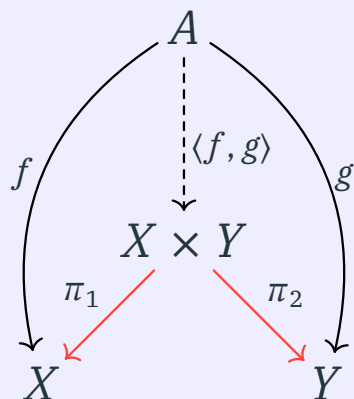
$$\Delta_X(x) = (x, x)$$

$$(f \times h)(a, b) =$$

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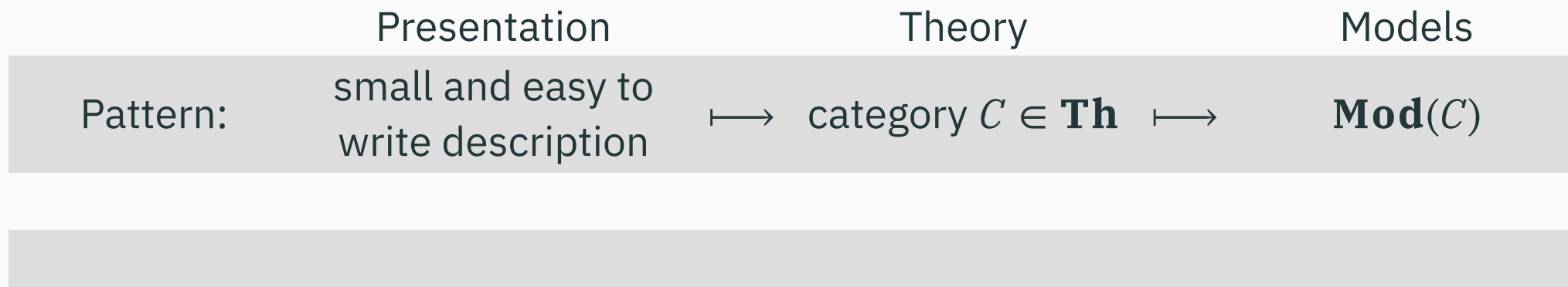
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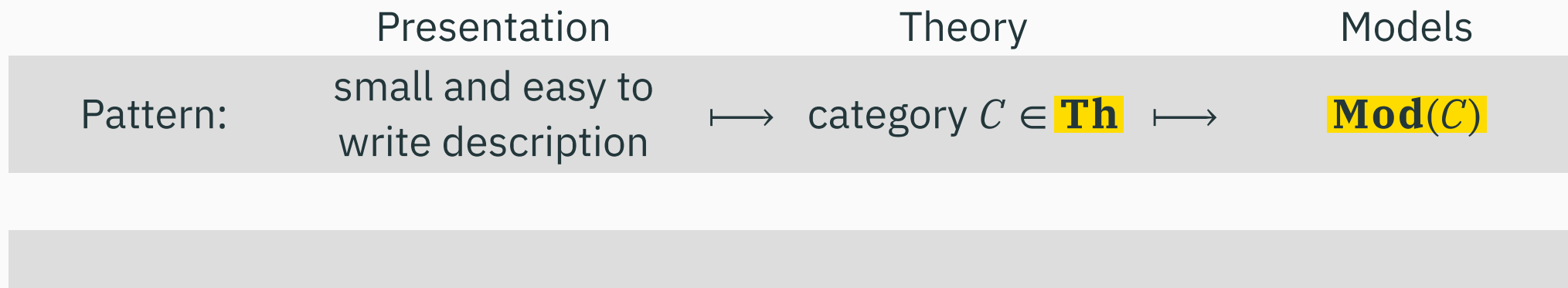
$$(f \times h)(a, b) = (f(a), h(b))$$

Initial-algebra semantics: usual pattern



1. Ehresmann C. 1965. *Catégories et Structures*. Dunod

Initial-algebra semantics: usual pattern



“Locally presentable categories”

$\hookrightarrow$ nice 👍 (in particular, have an initial object 😊)

Initial-algebra semantics: usual pattern

	Presentation		Theory		Models
Pattern:	small and easy to write description	$\mapsto$	category $C \in \mathbf{Th}$	$\mapsto$	$\mathbf{Mod}(C)$
Ehresmann ^[1] :	finite limit sketch	$\mapsto$	$C \in \mathbf{Lex}$	$\mapsto$	$\mathbf{LEX}(C, \mathbf{Set})$

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Lex = **l**eft **e**xact = categories with **finite limits** (products, terminal object).

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Let’s do an example!

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Example: theory of groups

Definition (Textbook definition of a group)

A group is a **set** G together with a **binary operation** on G , here denoted " $\cdot$ ", that combines any two **elements** a and b of G to form an element of G , denoted $a \cdot b$, such that the following three requirements, known as **group axioms**, are satisfied:^{[5][6][7][a]}

Associativity

For all a, b, c in G , one has $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Identity element

There exists an element e in G such that, for every a in G , one has $e \cdot a = a$ and $a \cdot e = a$.

Such an element e is unique. It is called the *identity element* (or sometimes *neutral element*) of the group.

Inverse element

For each a in G , there exists an element b in G such that $a \cdot b = e$ and $b \cdot a = e$, where e is the identity element.

For each a , the element b is unique; it is called *the inverse* of a and is commonly denoted a^{-1} .

Example $(\mathbb{Z}, +)$, $(\mathbb{R}^*, \times)$

Example: theory of groups

Definition (Finite limit sketch of groups)

- a sort X
- operations

$$\odot : X \times X \rightarrow X$$

$$e : 1 \rightarrow X$$

$$i : X \rightarrow X$$

- equations:

$$(x \odot y) \odot z = x \odot (y \odot z)$$

$$x \odot i(x) = e$$

$$e \odot x = x$$

$$\begin{array}{ccc} X \times X \times X & \xrightarrow{\odot \times id} & X \times X \\ id \times \odot \downarrow & & \downarrow \odot \\ X \times X & \xrightarrow{\odot} & X \end{array}$$

$$\begin{array}{ccccc} X & \xrightarrow{\Delta} & X \times X & \xrightarrow{id \times i} & X \times X \\ \downarrow ! & & & & \downarrow \odot \\ 1 & \xrightarrow{e} & & & X \end{array}$$

$$\begin{array}{ccccc} X & \xrightarrow{\cong} & 1 \times X & \xrightarrow{e \times id} & X \times X \\ & \searrow & & & \downarrow \odot \\ & & & & X \end{array}$$

+ $i(x) \odot x = e$ and $x \odot e = x$ 

Example: theory of groups

Definition (Theory of groups)

Category (with finite limits) C_{grp} generated from this sketch:

- $X, 1, X \times X, X \times X \times X, \dots \in \text{ob } C_{\text{grp}}$
- $\odot : X \times X \rightarrow X, i : X \rightarrow X, e : 1 \rightarrow X \in \text{mor } C_{\text{grp}}$
 $\Delta : X \rightarrow X \times X, ! : X \rightarrow 1, \text{id} \times e : X \times 1 \rightarrow X \times X, \dots \in \text{mor } C_{\text{grp}}$
 $i \circ e, \odot \circ \Delta, \dots \in \text{mor } C_{\text{grp}}$
- equations are satisfied

Example: theory of groups

Definition (Models of C_{grp})

$$\mathbf{Mod}(C_{\text{grp}}) := \mathbf{LEX}(C_{\text{grp}}, \mathbf{Set})$$

Finite limit preserving *functors* from C_{grp} to **Set**. A model $M : C_{\text{grp}} \rightarrow \mathbf{Set}$

- maps X to a **set** $G = M(X)$,
- maps $\odot : X \times X \rightarrow X$ to a **function** $+: G \times G \rightarrow G$,
- maps $e : 1 \rightarrow X$ to a **function** $\{*\} \rightarrow G$ i.e. a element $0 \in G$,
- maps $i : X \rightarrow X$ to a **function** $- : G \rightarrow G$,
- preserves the equations: e.g.

$$\begin{array}{ccc} G & \xrightarrow{\Delta} & G \times G \xrightarrow{id \times -} G \times G \\ \downarrow ! & & \downarrow + \\ \{*\} & \xrightarrow{"0"} & G \end{array}$$

$\Leftrightarrow$

$$+ \circ (id \times -) \circ \Delta = 0$$

$$\forall g \in G, g + -g = 0$$

...

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$$\begin{array}{ccc} G & \xrightarrow{\Delta} & G \times G \xrightarrow{\text{id} \times -} G \times G \\ \downarrow ! & & \downarrow + \\ \{*\} & \xrightarrow{\text{"0"}} & G \end{array} \quad \Leftrightarrow \quad \begin{array}{l} + \circ (\text{id} \times -) \circ \Delta = 0 \\ \forall g \in G, g + -g = 0 \end{array}$$

...

Example: theory of groups

Remark (Model morphism)

Model morphism = function between the sets that preserves the operations

↳ *group homomorphism*

Remark (Initial group)

Trivial group $G = \{0\}$

(Not a very interesting programming language...)

Conclusion

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My work:	Skel program	$\mapsto$	$\mathbb{C} \in \mathbf{VDbTh}$	$\mapsto$	$\mathbf{VDBTH}(\mathbb{C}, \mathbf{Rel})$

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Appendix

- [1] C. Ehresmann, *Catégories et Structures*. Dunod, 1965.